

Bank Customer Churn Data Analytics (DA) Report

1. Problem Statement

The banking business wants to analyze its historical customer data to understand overall performance, identify customer churn patterns, evaluate regional and demographic trends, and support data-driven decision-making. The main objective is to improve customer retention by optimizing customer engagement, identifying risk factors for churn across different segments, and monitoring key financial metrics.

2. Dataset Overview

The Bank Customer Churn dataset contains transactional and demographic records with the following key fields:

- **Customer Details:** Customer ID, Surname, Age, Gender
 - **Geographic Details:** Geography, Country
 - **Account & Financial Details:** Credit Score, Tenure, Balance, Estimated Salary, Num of Products
 - **Activity Metrics:** Has Credit Card, Is Active Member, Exited (Churn Status)
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3. Data Preprocessing Steps

The following preprocessing steps were performed before creating the report:

- Removed duplicate and redundant records
 - Checked and handled null or blank values
 - Standardized column names and data types for consistency
 - Created necessary conditional columns and categories for age groups and activity status
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4. Sheets / Pages Created in the Report

4.1 Executive Overview Dashboard

Purpose: High-level view of overall bank and customer performance.

Key Components: Total Customers, Total Churned Customers, Active Customers, Churn Rate %, Average Credit Score, Total Balance, and regional distribution.

4.2 Customer Demographics & Behavior

Purpose: Understand customer demographics and behavior contribution.

Key Components: Total Male Customers, Total Female Customers, Active Members, Credit Card Holders, Avg Balance of Churned customers, Age distribution, and Gender-wise churn analysis.

4.3 Financial & Activity Analysis

Purpose: Evaluate the financial health and activity status of customers.

Key Components: Total Balance, Average Balance, Avg Estimated Salary, Active Members %, Total Churned Balance, Product-wise balance distribution, and Tenure analysis.

5. DAX Measures Used

5.1 Basic Measures

- Total Balance = SUM(Churn_Modelling[Balance])
- Average Balance = AVERAGE(Churn_Modelling[Balance])
- Avg Estimated Salary = AVERAGE(Churn_Modelling[EstimatedSalary])

5.2 Derived Measures

- Churn Rate % = $\text{DIVIDE}(\text{CALCULATE}(\text{COUNT}(\text{Churn_Modelling}[\text{Exited}]), \text{Churn_Modelling}[\text{Exited}] = 1), \text{COUNT}(\text{Churn_Modelling}[\text{CustomerId}], 0) * 100$
 - Active Members % = $\text{DIVIDE}(\text{CALCULATE}(\text{COUNT}(\text{Churn_Modelling}[\text{CustomerId}], \text{Churn_Modelling}[\text{IsActiveMember}] = 1), \text{COUNT}(\text{Churn_Modelling}[\text{CustomerId}], 0) * 100$
 - Total Churned Balance = $\text{CALCULATE}(\text{SUM}(\text{Churn_Modelling}[\text{Balance}]), \text{Churn_Modelling}[\text{Exited}] = 1)$
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6. Key Insights & Findings

- Customers within specific age brackets exhibit higher churn rates compared to other groups.
 - Inactive members show a significantly higher tendency to exit the bank.
 - Geographic distribution reveals variations in customer retention across different regions.
 - A substantial amount of total customer balance is held by accounts that eventually get churned.
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7. Final Assumptions & Business Recommendations

Assumptions

- The data provided represents accurate historical banking and customer records.
- External macroeconomic factors were not factored into individual churn predictions.
- Account transaction history beyond the scope of current attributes is excluded.

Recommendations

- Implement targeted retention campaigns for inactive members to boost engagement.

- Focus special customer service initiatives on high-balance customer segments to prevent high-value churn.
- Review product offerings for customer segments showing elevated churn probabilities.

8. Final Conclusion (DA Report Summary)

The Bank Customer Churn Data Analytics report provides actionable insights into customer retention drivers, demographic behavior, and financial exposure. By leveraging interactive dashboards and DAX-driven measures, the business can identify at-risk customer segments and growth opportunities. Implementing these recommended strategies can significantly improve customer lifetime value and overall operational efficiency.